Admin Panel - Full Specification

Dwarka Dental Clinic Management System | Document 1 of 3 | Control tower of the clinic

1. Purpose of the Admin panel

The Admin is the clinic owner / manager. This panel must give complete control over people (staff, doctors, salary, attendance), money (revenue, expenses, profit), operations (calendar, inventory, lab) and security (roles, audit). Every number the Admin sees must come automatically from work done in the Receptionist and Doctor panels - nothing should be typed twice.

How to read this document: every feature is tagged EXISTS (already in the README), PARTIAL (exists but incomplete) or NEW (not in the system yet). Receptionist and Doctor documents follow the same format, and Section 7 shows how all three panels connect.

2. What the system has today (Admin)

Area	Currently available	Problem / gap
Dashboard	Patients, today's appointments, missed / upcoming, revenue (today + total), appointments by category.	Counts only. No trends, dues, no-show rate, doctor-wise or period filters.
Patients	Searchable list, status filter, full detail with clinical + payments + AI reports.	No merge duplicates, export, document center, or consent tracking.
Appointments	Filters today / upcoming / missed / all, by doctor, status timeline.	No master calendar, chairs / rooms, holidays, waiting list or no-show analytics.
Treatment categories	CRUD with default duration and follow-up days.	No price list, tax, packages or procedure-level pricing.
Staff	Create user, set password, activate / deactivate (5 roles).	No profile, documents, attendance, leave, shifts, salary, payroll or permission control.
Revenue	Payments list, summary by method, invoice / receipt tracking.	Income only. No expenses, refunds, discounts approval, dues ageing or profit.
Reports / Settings	Pages are listed in the README.	No defined reports or settings. Treat as placeholders to build.
Notifications	WhatsApp reminder dispatch only.	No in-app alert / approval system between panels (parked for now).
Security	JWT + role guard, bcrypt, RLS, audit_logs table.	No audit viewer, login history, lockout, refresh tokens, 2FA. Many endpoints accept any valid role.

Technical issues to fix first

- doctor_availability table exists but no API / screen uses it, so bookings ignore working hours, leave and holidays.
- audit_logs table exists but nothing reads it; confirm every create / update / delete writes a before / after entry.
- treatment_plans is only a stub; billing cannot be driven by a doctor's plan yet.
- Authorization is coarse: GET /patients, /payments and POST /categories accept any (or several) roles. Tighten to the matrix in Section 6.
- Legacy Mongoose models are dead code - delete them. Disable auto-seed and demo passwords in production.
- JWT in localStorage with 8h expiry and no refresh: add refresh tokens, login rate-limit and account lockout.

3. Admin modules - detailed specification

A1 Dashboard and Analytics

Goal: One screen answering: how is the clinic doing today, this week, this month, and what needs my attention?

Feature	Status	What it must do
KPI cards	PARTIAL	Keep existing cards. Add outstanding dues, new vs returning patients, no-show %, average bill, chair utilization, collection vs target.
Date range filter	NEW	Today / 7 days / month / custom. Every widget obeys it. Compare with previous period.
Revenue and expense trend	NEW	Line chart of income vs expense with profit line, by day / week / month.
Doctor-wise performance	NEW	Patients seen, procedures, revenue, average consultation time, follow-up conversion per doctor.
Action center	NEW	Pending approvals (discount, refund, leave, expense), low stock, expiring items, unreviewed AI reports, overdue lab work.
Live clinic board	NEW	Who is waiting, who is in chair, which doctor is free, today's attendance (present / absent / on leave).
Category breakdown	EXISTS	Appointments by treatment category; add revenue per category.

Linked with: Receptionist (queue, collections), Doctor (procedures), HR (attendance), Inventory, Finance.

A2 Staff and HR Management

Goal: Complete employee lifecycle from joining to payroll, in one place.

Feature	Status	What it must do
Staff accounts	EXISTS	Create, set password, activate / deactivate. Add: archive (never hard delete), force password reset, unlock account.
Employee profile	NEW	Employee ID, photo, designation, department, joining date, probation, emergency contact, address, ID proofs, bank details, qualification, registration no. (doctors).
Staff documents	NEW	Upload appointment letter, ID, degree, license; expiry alerts for council registration and contracts.
Shifts and roster	NEW	Define shifts (morning / evening / split), weekly roster per staff, week-off, swap requests, roster published to staff panels.
Attendance	NEW	Daily check-in / check-out (self-service from panel, with admin manual correction and reason). Late, early-out, half-day, absent, overtime auto-marked against roster. Monthly attendance register and export.
Leave management	NEW	Leave types (casual, sick, paid, unpaid), yearly balance, apply from panel, admin approve / reject, auto-block appointments for doctors on approved leave, calendar of who is off.
Salary structure	NEW	Per employee: basic, allowances, fixed deductions. Doctors: fixed salary, per-visit fee, or percentage share per procedure / category (links to A3).
Payroll run	NEW	Monthly run pulls attendance, unpaid leave, overtime, advances, incentives and commission; shows preview; admin locks and marks paid with method and date.
Advances and loans	NEW	Record advance, set monthly recovery, auto-deduct in payroll, show balance.
Payslips	NEW	PDF payslip per month; staff can view / download only their own.
Role and permission matrix	NEW	Roles: admin, doctor, receptionist, assistant, lab technician. Admin can toggle permissions per role (Section 6) and override per user.
Staff performance	NEW	Attendance score, tasks, patient feedback, doctor KPIs; notes and warnings log per employee.

Linked with: Attendance and leave (Receptionist and Doctor self-service), Appointments (doctor leave blocks), Finance (salary expense), Audit.

A3 Doctor Management

Goal: Control each doctor's availability, fees and earnings model.

Feature	Status	What it must do
Doctor list	EXISTS	Active doctors with specialization and contact.
Weekly availability	PARTIAL	Build UI + API on doctor_availability: working days, start / end, break, slot length, chair, max patients per day.
Leave and blocked slots	NEW	Doctor leave, conferences, emergency blocks; booking screen respects them.
Fees	NEW	Consultation fee, follow-up fee, procedure-specific fee overrides.
Revenue share / commission	NEW	Percentage or fixed per treatment category, lab-cost deduction rule, monthly doctor statement feeding payroll.
Doctor report card	NEW	Procedures, revenue, repeat patients, cancellations, average chair time.

Linked with: Appointments (slots), Billing (fee defaults), Payroll (commission), Doctor panel (schedule).

A4 Patient Administration

Goal: Admin oversight of the patient database, data quality and privacy.

Feature	Status	What it must do
Patient list and detail	EXISTS	Search, status filters, full profile and history.
Duplicate detection and merge	NEW	Flag same phone / name+DOB; merge keeps all appointments, records, payments, files under one patient number.
Reassign doctor	NEW	Bulk or single reassignment when a doctor leaves; history retained.
Archive / restore	PARTIAL	Soft-delete exists; add reason, restore, and list of archived patients.
Document center	NEW	All uploads per patient: X-rays, reports, consent forms, ID. Type, date, uploader, preview.
Consent and privacy	NEW	Consent forms signed / pending per patient; record who viewed sensitive data.
Export	NEW	CSV / PDF export of patient list and single patient summary.

Linked with: Receptionist (registration), Doctor (clinical data), Audit log.

A5 Appointment and Calendar Control

Goal: Master view of the clinic schedule with rules the front desk must follow.

Feature	Status	What it must do
Appointment list and timeline	EXISTS	Filters, per doctor, status flow scheduled > confirmed > arrived > in-progress > completed.
Master calendar	NEW	Day / week view, column per doctor or chair, drag to reschedule, colour by status.
Chairs and rooms	NEW	Define chairs / operatories; a booking reserves doctor AND chair; conflict check covers both.
Working hours and holidays	NEW	Clinic hours, public holidays, closures; blocks booking automatically.
Booking rules	NEW	Buffer time, overbooking limit, advance-booking window, emergency slots, walk-in allowance.
Cancel / no-show reasons	NEW	Mandatory reason codes; no-show and cancellation analytics by patient, doctor, reason.
Waiting list	NEW	Patients waiting for an earlier slot; suggest when a cancellation frees one.
Override	NEW	Admin can override conflict with a reason, logged to audit.

Linked with: Receptionist booking, Doctor availability, Leave, Settings.

A6 Treatment Catalog and Pricing

Goal: A single source of truth for what the clinic offers and charges.

Feature	Status	What it must do
Treatment categories	EXISTS	CRUD, default duration, follow-up days, active toggle. Restrict create / edit to Admin only.
Procedure price list	NEW	Procedures under each category with price, tax, duration, consumables used, doctor share.
Packages and plans	NEW	Bundles (e.g. whitening + cleaning), validity, installment option.
Discount rules	NEW	Reception discount limit (e.g. up to 5%), above that needs approval; senior citizen / staff / referral discounts.
Treatment plan template	NEW	Standard multi-visit plans doctors can pick (e.g. RCT + crown) with estimated cost.

Linked with: Doctor treatment planning, Receptionist billing, Inventory consumption.

A7 Billing, Revenue and Finance

Goal: Know income, expenses and profit with an auditable trail.

Feature	Status	What it must do
Payments and summary	EXISTS	Payment list, summary by cash / UPI / card / bank, invoices and receipts.
Invoice management	PARTIAL	View, void / cancel with reason, credit note instead of deleting, numbering prefix from settings.
Refunds	NEW	Receptionist raises request, Admin approves, refund recorded against invoice and method.
Discount approvals	NEW	Queue of discounts above the limit with patient, amount, reason, requester.
Outstanding dues	NEW	Dues per patient with ageing (0-30, 31-60, 60+ days), reminders list for front desk.
Installments / treatment-linked billing	NEW	Plan-wise installment schedule, auto-link payments to treatment plan stages.
Expense management	NEW	Categories (rent, salary, lab, consumables, utilities, marketing, maintenance), vendor, bill upload, payment method, recurring expenses.
Daily cash closing	NEW	Receptionist submits day-end cash count, system compares with recorded cash receipts, Admin sees difference and signs off.
Profit and loss	NEW	Income minus expenses minus salary by month; doctor-wise and category-wise profitability.
Tax / GST reports	NEW	Tax summary by period if applicable; export for accountant.

Linked with: Receptionist billing, Payroll, Inventory purchases, Lab bills, Reports.

A8 Inventory, Lab and Equipment

Goal: Stop stock-outs and track outsourced lab work.

Feature	Status	What it must do
Item master	NEW	Materials and consumables with unit, category, reorder level, supplier.
Stock in / out	NEW	Purchase entry, issue to clinic, auto-deduct per procedure, wastage and adjustment with reason.
Alerts	NEW	Low stock and expiry (30 / 60 / 90 days) shown in Action Center.
Suppliers and purchase orders	NEW	Vendor list, PO creation, receive against PO, vendor payment status.
Lab orders	NEW	Crown, denture, aligner orders raised by doctor: lab name, shade, due date, status (sent / received / fitted), cost, linked to patient and invoice.
Equipment and sterilization log	NEW	Equipment list, service due dates, daily autoclave log, maintenance history.

Linked with: Doctor (lab orders, consumables), Receptionist (stock requests, lab delivery), Finance.

A9 Reports Center

Goal: Ready-made, exportable reports. The current Reports page is a placeholder - build the following.

Feature	Status	What it must do
Financial	NEW	Daily collection, monthly revenue, payment-method split, dues ageing, expenses, P&L, doctor payout.
Patients	NEW	New vs returning, source of referral, age / gender, inactive patients, retention.
Appointments	NEW	Utilization, no-show / cancellation, peak hours, average wait, follow-up compliance.
Clinical	NEW	Procedures performed, diagnosis frequency, AI X-ray accuracy (confirmed vs rejected).
HR	NEW	Attendance register, leave summary, payroll register, overtime.
Inventory	NEW	Stock valuation, consumption by procedure, expiry, purchases by vendor.
Export and schedule	NEW	PDF / Excel export on every report; saved filters.

Linked with: All modules.

A10 Audit, Security and Data Protection

Goal: Every sensitive action is traceable; access is least-privilege.

Feature	Status	What it must do
Audit log viewer	NEW	Search by user, module, action, date; show before / after JSON. Immutable.
Login history and sessions	NEW	IP, device, time, failures; admin can end sessions.
Account protection	NEW	Lockout after repeated failures, rate-limit, password policy, optional 2FA for Admin and Doctor, refresh tokens.
Backup and export	NEW	Scheduled database backup confirmation, manual full export, restore procedure documented.
Sensitive-access log	NEW	Log who opened which patient's clinical data; doctors viewing patients outside their list must give a reason.

Linked with: Every panel.

A11 Clinic Settings

Goal: All configurable values in one place so nothing is hard-coded. The current Settings page is a placeholder.

Feature	Status	What it must do
Clinic profile	NEW	Name, logo, address, phone, registration / GST numbers, signature image.
Hours and holidays	NEW	Weekly hours, holiday calendar (feeds A5).
Numbering	NEW	Prefix and format for patient, appointment, invoice and receipt numbers.
Print templates	NEW	Invoice, receipt, prescription header / footer, certificate, consent form templates.
Payment and tax	NEW	Enabled methods, tax rates, currency format, discount limits per role.
Defaults	NEW	Appointment duration, follow-up days, reminder timing (WhatsApp later), session timeout.
Master data	NEW	Referral sources, cancellation reasons, expense categories, leave types, lab vendors.

Linked with: Every panel reads these values.

A12 Internal Notifications and Tasks

Goal: In-app only for now (WhatsApp is parked). Uses the existing notifications table.

Feature	Status	What it must do
Notification center	PARTIAL	Bell with unread count in all panels; click opens the related record.
Approval requests	NEW	Discount, refund, leave, expense, stock request routed to Admin with approve / reject and comment.

Feature	Status	What it must do
Announcements	NEW	Admin posts notice to all staff or one role; read receipts.
Task assignment	NEW	Admin assigns task to a staff member with due date; staff marks done.

Linked with: Receptionist and Doctor panels.

4. Admin database additions

Existing 18 tables stay. Add the following (all with clinic_id, created_at, updated_at, soft-delete, and audit logging).

Table	Purpose	Key fields
staff_profiles	Employee details	user_id, employee_code, designation, join_date, bank, documents, emergency contact
shifts / rosters	Shift definitions and assignments	shift name, start, end, user_id, date, week_off
attendance	Daily attendance	user_id, date, check_in, check_out, status, source, corrected_by, reason
leave_types / leave_balances / leave_requests	Leave system	type, days, from, to, status, approver
salary_structures	Pay components per staff	basic, allowances, deductions, commission_rule
payroll_runs / payslips	Monthly payroll	month, status, gross, deductions, net, paid_on, method
staff_advances	Advances and recovery	amount, monthly_recovery, balance
permissions / role_permissions / user_overrides	Configurable access	module, action, role, allowed
chairs	Operatories	name, status
clinic_holidays	Closures	date, reason
procedures / price_list / packages	Pricing	category_id, price, tax, duration, doctor_share
discount_rules / refunds / credit_notes	Billing control	limit, approver, invoice_id, amount, reason
expenses / expense_categories / vendors	Cost side	amount, date, vendor, bill file, recurring flag
cash_closings	Day-end reconciliation	date, expected, counted, difference, signed_off_by
inventory_items / stock_movements / purchase_orders	Stock	item, qty, type, reorder_level, expiry, vendor
lab_orders	Outsourced lab work	patient_id, doctor_id, type, vendor, due, status, cost
equipment / sterilization_logs	Assets and hygiene	service_due, cycle, operator
tasks / announcements	Internal coordination	assignee, due, status, audience
clinic_settings	Key-value settings	key, value, updated_by

5. Admin API additions (summary)

Group	Endpoints to add
HR	/api/staff/:id/profile, /api/attendance (check-in, check-out, correct, monthly), /api/leaves (apply, approve), /api/shifts, /api/payroll (preview, run, lock, pay), /api/payslips/:id, /api/advances
Access	/api/roles, /api/permissions, /api/audit-logs (search), /api/sessions, /api/auth/refresh, /api/auth/logout
Scheduling	/api/availability, /api/chairs, /api/holidays, /api/waiting-list, /api/appointments/:id/override
Pricing	/api/procedures, /api/price-list, /api/packages, /api/discount-rules
Finance	/api/invoices/:id/void, /api/refunds, /api/approvals, /api/dues, /api/expenses, /api/vendors, /api/cash-closing, /api/finance/pnl
Inventory and lab	/api/inventory, /api/stock-movements, /api/purchase-orders, /api/lab-orders, /api/equipment

Group	Endpoints to add
Platform	/api/reports/:type (with export), /api/settings, /api/notifications, /api/tasks, /api/announcements, /api/patients/merge

6. Permission matrix (whole system)

This matrix is the rule set for all three panels. Implement it once in the backend as role + permission checks, and hide controls in the UI that the role cannot use.

Capability	Admin	Receptionist	Doctor
Register / edit patient demographics	Full	Full (no delete)	View + edit medical history
Delete / archive / merge patient	Full	None	None
Clinical records, dental chart, diagnosis	View + audit	None (summary only)	Full (own patients)
Prescriptions	View	Print only	Create / print
Book / reschedule / cancel appointments	Full	Full	Own schedule: confirm, reschedule request
Check-in (arrived) and queue	Full	Full	View live queue
Start / complete treatment	Override	None	Full
Invoices and payments	Full + approve	Create / collect	View own procedures
Discounts and refunds	Approve	Request (limit applies)	None
Revenue and finance reports	Full	Own collections only	Own earnings only
Expenses, inventory, vendors	Full	Raise stock request	Raise consumable request
Staff profiles, salary, payroll	Full	Own payslip only	Own payslip only
Attendance and leave	Full + approve	Self check-in / apply	Self check-in / apply
Treatment catalog and price list	Full	View	View + suggest
Clinic settings, roles, permissions	Full	None	None
Audit log and login history	Full	None	Own activity only
AI X-ray upload / review	View	Upload	Upload + confirm / reject

7. How the Admin panel connects to the other two

When this happens	Admin panel automatically gets
Receptionist registers patient or books appointment	Patient count, calendar, utilization and referral-source stats update.
Receptionist checks patient in	Live clinic board shows waiting patient; average wait time recorded.
Doctor completes treatment and records procedures	Revenue-by-category, doctor performance, inventory auto-deduction, commission accrual.
Receptionist records payment / requests discount or refund	Revenue, cash closing, dues; approval request appears in Action Center.
Staff checks in / applies for leave	Attendance register and leave queue; doctor leave blocks the calendar.
Doctor raises lab order or consumable request	Lab board and stock request queue.
Admin changes price, hours, holiday, permission	Receptionist booking / billing and Doctor schedule use the new values immediately.
Any create / update / delete	Audit log entry with user, time and before / after.

8. Build phases for the Admin panel

Phase	Scope	Result
1 Foundation	Fix security gaps, permission matrix, audit viewer, clinic settings, delete dead code, doctor availability + holidays + chairs	Safe base that every other panel can rely on
2 HR	Staff profiles, shifts, attendance, leave, salary structure, payroll, payslips, advances	Complete staff management
3 Finance	Price list, discount rules, approvals, refunds, dues, expenses, cash closing, P&L	Income, expense and profit visibility
4 Operations	Inventory, purchase orders, lab orders, equipment, patient merge / documents	Supply chain and lab under control
5 Insight	Dashboard v2, Reports center with export, internal tasks and announcements	Decision-ready analytics

9. Admin acceptance checklist

- Admin can create an employee, set shift, see their attendance for the month and run payroll with correct net pay.
- A doctor on approved leave cannot be booked by the Receptionist.
- A discount above the limit cannot be applied until Admin approves.
- Dashboard revenue equals the sum of payments, with no manual entry anywhere.
- Cash closing shows expected vs counted difference per day.
- Every edit to patient, invoice, payment or staff record appears in the audit log.
- Receptionist and Doctor roles are blocked by the backend (not just hidden in the UI) from Admin-only actions.
- Every report exports to PDF or Excel.